Learn More and Learn Usefully: Truncation Compensation Network for Semantic Segmentation of High-Resolution Remote Sensing Images

Li Zhang, Zhenshan Tan, Guo Zhang, Wen Zhang, and Zhijiang Li

***Abstract*— Semantic segmentation of high-resolution remote-sensing images (HR-RSIs) focuses on classifying each pixel of input images. Recent methods have incorporated a downscaled global image as supplementary input to alleviate global context loss from cropping. Nonetheless, these methods encounter two key challenges: diminished detail in features due to down-sampling of the global auxiliary image (GAI) and noise from the same image that reduces the network's discriminability of useful and useless information. To overcome these challenges, we propose a truncation compensation network (TCNet) for HR-RSI semantic segmentation. TCNet features three pivotal modules: the guidance feature extraction module (GFM), the related-category semantic enhancement module (RSEM), and the global–local contextual cross-fusion module (CFM). GFM focuses on compensating for truncated features in the local image and minimizing noise to emphasize learning of useful information. RSEM enhances discernment of global semantic information by predicting spatial positions of related categories and establishing spatial mappings for each. CFM facilitates local image semantic segmentation with extensive contextual information by transferring information from global to local feature maps. Extensive testing on the ISPRS, BLU, and GID datasets confirms the superior efficiency of TCNet over other approaches.**

***Index Terms*— Context fusion, remote-sensing images, semantic enhancement, semantic segmentation, useful information.**

I. Introduction

SEMANTIC segmentation of high-resolution remote-sensing images (HR-RSIs) is designed to classify input images pixel-by-pixel, using key ground information (e.g., satellite images to identify buildings, water, vegetation, roads) This improves our understanding and utilization of HR-RSIs, with widespread applications in fields such as land and resource planning [1], urban monitoring [2], disaster assessment [3], [4], and ecological environment monitoring [5].

Contextual information is critical in semantic segmentation [6]. Techniques such as ParseNet [7] have enhanced pixel feature representation by incorporating contextual information, thus refining the semantic interpretation of the network. Some methods leverage this information to encourage networks to consider the surrounding objects during classification, reducing misclassification risks [8], [9], [10]. Contextual information in this context encapsulates both the relationships among individual entities and their connections to the overall image [11], [12]. However, the wide scope of HR-RSIs challenges the use of entire images in deep convolutional neural networks (DCNNs). Commonly, images are cropped before being inputted into networks, but this truncates some complete features, resulting in a loss of global contextual information [13], [14]. To counter this, researchers have proposed two main strategies. The first aggregates as much contextual information from the input image as possible [15], [16], focusing mainly on local context and neglecting the compensation for truncated features in large-scale HR-RSIs. The second, multi-field-of-view (MFOV) image fusion, uses low-resolution global auxiliary images (GAIs) to capture broad spatial contexts and complete contours [as illustrated in Fig. 1(a)], along with high-resolution local images for detailed information [17], [18], [19], [20], [21]. Surprisingly, we found that GAIs might sometimes degrade network performance, especially in complex textures. As demonstrated in Fig. 1(d), using the Potsdam dataset, we trained two models: a fully convolutional network (FCN) [22] and a dual-branch network with GAI. The ordinate of Fig. 1(d) the dual-branch network's performance enhancement with GAI, compared with the standalone FCN, varied across different images. It can be observed from Fig. 1(d), while GAI restored accuracy in most cases, some images showed a decrease in semantic segmentation accuracy after GAI integration, such as the second and seventh images. This discrepancy can be attributed to two factors. First, inputting an entire GAI into the network brings in a

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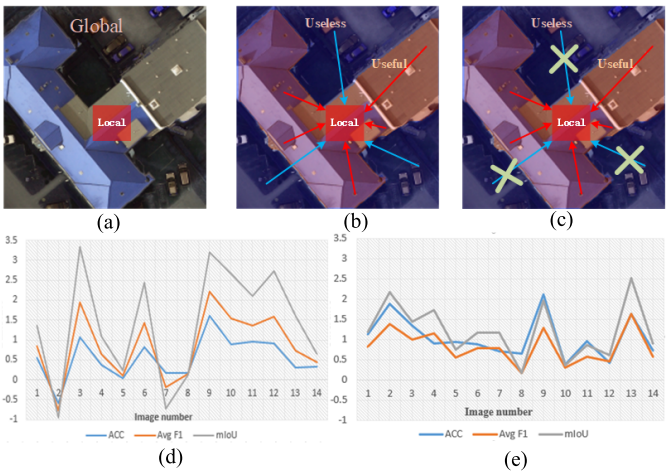

Fig. 1. (a) GAI. "Local" is the position of local segmented image in GAI. (b) Previous method takes GAI as whole input, and image contains both useful and useless information. Useful information: regions linked to local segmented image. Useless information: regions unrelated to local segmented image. (c) GUAI. Compared with GAI, GUAI masks out useless areas in GAI. (d) Experimental results with GAI; abscissa represents different image names, and ordinate represents improvement effect after adding GAI images. (e) Experimental results with GUAI; ordinate represents improvement effect after adding GUAI images.

large amount of irrelevant background information. Second, the downsampling of GAI might lead to a loss of intricate details, complicating the task of learning accurate global feature mapping.

To tackle these challenges, we designed a truncation compensation network (TCNet) to extract useful information. We categorize regions linked to the local image as useful, while those unrelated are considered useless, as shown in Fig. 1(b). We refer to a GAI containing only useful information as GUAI, as shown in Fig. 1(c). TCNet includes three key modules: an explicit guidance feature extraction module (GFM), a related-category semantic enhancement module (RSEM), and a contextual cross-fusion module (CFM). The GFM effectively filters out useless information, guiding the network to focus on useful context to make up for the truncated features in the local image. As shown in the comparison between Fig. 1(d) and (e), integrating GFM significantly diminishes the impact of irrelevant information. This selective removal of interference substantially enhances the ability of the network to discriminate between useful and useless information in GAI. To overcome the loss of spatial details in the downsampled GUAI, we introduced RSEM. This module aggregates semantic information from pixels within related categories, enhancing the discriminability of category-specific information. It compensates for the loss of detail caused by downsampling, thereby bolstering feature enhancement. Finally, we devised the CFM to establish a connection between local and global contexts. This module maps global features onto local features, aiding in the final segmentation process. The main contributions of our study are as follows.

1) We explore the auxiliary role of GAI in providing global context for segmentation and investigate the information that positively influences local image segmentation.
2) We introduce an explicit GFM to filter GAI for valuable features that enhance global context provision, thereby mitigating the detrimental impact of irrelevant features on HR-RSIs' semantic segmentation.
3) We propose an RSEM to minimize semantic discrepancies between pixels of related categories. This improves the ability of the network to distinguish between similar categories, thereby enhancing segmentation performance.
4) We introduce a global–local contextual CFM, designed to map global features to local features. This compensates for the truncated features in local images, enriching the overall segmentation process.

II. Related Works

In recent years, DCNNs have made remarkable advancements in image semantic segmentation, as evidenced by various studies [23], [24]. FCNs, such as FCN [22], U-Net [25], Segnet [26], and DeepLab [27], [28], [29], [30], have become predominant in the semantic segmentation of HR-RSIs. Unlike traditional machine learning methods, DCNN-based methods excel at capturing detailed contextual information, enhancing their feature representation and pattern recognition capabilities [31], [32], [33]. The effective aggregation of global context information is crucial in semantic segmentation, especially for natural images. Integrated with the prevalent methods of aggregation context in semantic segmentation of remote-sensing images, we provide a summary of semantic segmentation from two perspectives.

A. Single-Branch Network for Semantic Segmentation With Context Aggregation

In the realm of semantic segmentation for natural images, the effective aggregation of global context information is significant. Liu et al. [7] introduced ParseNet, which uses diverse levels of context to resolve local confusion. This network aggregates image contexts through global pooling, combining them with original feature maps through early or late fusion. Chen et al. [28] developed DeepLab, which features an atrous spatial pyramid pooling (ASPP) module. This approach, replacing parts of the downsampling layer, aggregates contexts from different receptive fields without drastically reducing the resolution of feature maps. This technique enables the effective aggregation of global context into local information. Zhao et al. [34] proposed PSPNet to enhance network proficiency in comprehending global context. It uses a pyramid pooling module that aggregates context from various regions, augmenting the capability of the network to capture context from the entire images.

Building upon the success in semantic segmentation of natural scenes, various methods have been developed to aggregate contextual information in the semantic segmentation of HR-RSIs. Liu et al. [35] introduced an hourglass-shaped network for this purpose, with an encoder–decoder style that differs from the U-Net backbone by not using all the spatial context features of the encoder in the decoder branch. Diakogiannis et al. [36] proposed the ResUNet network framework, which combines the U-Net encoder/decoder

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backbone with residual connections, atrous convolutions, pyramidal scene parsing pooling, and multitask inference for extensive context aggregation. Drawing inspiration from the Swin transfer global modeling capabilities, He et al. [16] developed the ST-UNet network. This design embeds the Swin Transformer as an auxiliary encoder in the classic U-Net network, with a relational aggregation module to extract context from the auxiliary encoder and guide the main encoder. Zhang et al. [37] proposed the SwinB−CNN+ $BD_{1,2,3}$ for very high-resolution (VHR) remote-sensing image segmentation. This architecture merges the strengths of the Swin Transformer with the CNN-based decoders and includes an ASPP block based on depthwise separable convolution (SASPP) to capture multiscale contextual information. Wang et al. [31] designed a U-Net-like network structure using a transformer decoder with an efficient global and local attention mechanism to simulate both types of information in the decoder. GLOTS [38] represents a global–local transformer segmentation framework that uses a transformer structure for both the encoding and decoding input data, effectively using global and local features to obtain consistent feature representation and segmentation of remote-sensing images.

However, these methods often overly emphasize local details, neglecting the broader contexts crucial in large-scale remote-sensing images. Consequently, their application to the semantic segmentation of HR-RSIs can be limited.

B. Dual-Branch Network for Semantic Segmentation With Context Aggregation

Numerous methods use dual-branch networks to concurrently extract global and local context information [39] from diverse scenarios. Poudel et al. [40] propose ContextNet, a dual-resolution branch network based on a feature pyramid. In this network, the low-resolution branch is responsible for providing accurate labels, while the high-resolution branch refines the segmentation results. Similarly, Yu et al. [41] proposed a bilateral semantic segmentation network, which uses a spatial branch to preserve spatial details alongside a context branch for capturing global context information. These upper and lower branches are then fused to improve segmentation accuracy while maintaining processing speed. Mazzini [42] developed the guided upsampling network (GUN), a multiresolution network that combines deep and shallow branches. This design efficiently leverages high-resolution details during the upsampling process. Other methods, such as DANet [43], CF-Net [44], and OCNet [45], focus on the relationship between a location pixel and its surrounding context pixels. They prioritize aggregating similar context pixel representations with higher weights, enhancing the contextual understanding. Yuan et al. [46] approached the issue using contextual features of the corresponding object pixels for feature enhancement. Their method involves aggregating the contextual features within each object area, thus strengthening the feature representation of individual pixels.

Zhou et al. [47] proposed the hierarchical context network (HCNet), which learns contextual information by exploring pixel-to-pixel (P2P) and pixel-to-object (P2O) relationships. While P2P describes the spatial details of the object, P2O provides semantic information, albeit not on a global scale. Other researchers have emphasized global context information in complex scenarios, such as Chen et al. [21] proposed the collaborative global–local networks (GLNet) extract both local and global information through dual branches, capturing high-resolution structures from enlarged local patches and contextual dependencies from low-sampled inputs. However, this approach ignores the varying importance of different image patches, treating them equally and integrating them with the entire image, resulting in high computational costs. To solve this problem, Wu et al. [48] introduced a patch-suggestion network that adaptively distinguishes the importance of different patches, selecting important ones over trivial ones. This method effectively alleviates the burden of excessive patches, though it still falls short in perfectly filtering useful information. Similarly, Ding et al. [20] used downsampled images to broaden the overall receptive field, introducing local feature adjustments for targeting boundaries and fusing information learned at multiple levels for better prediction accuracy. Ding et al. [49] proposed a deep learning network architecture with a two-stage multiscale training strategy. The first stage uses downsampled images to enlarge the receptive field and capture more contextual information, improving tolerance to inaccurate reference data. The second stage introduces local features to refine the prediction of object boundaries.

In the broader context of semantic segmentation of HR-RSIs, the existing methods use a variety of techniques aimed at augmenting the receptive field or integrating global context into local image segmentation. However, models that excessively focus on local details may fail to capture broader contexts, while those emphasizing global context might miss fine-grained details. Notably, few researchers have concentrated on distinguishing between useful and useless contexts in global remote-sensing images, a crucial aspect for effective semantic segmentation. Unfortunately, this gap often results in unsatisfactory performance, particularly under truncated conditions.

III. Methods

A. Network Architecture

In our study, a high-resolution, wide field-of-view image pair (I, L) is cropped to generate a local image pair (x^s, label^s) for network input. Simultaneously, to capture global contextual information, (I, L) is downsampled to create GAI represented as (x^l, label^l). Both (x^s, label^s) and (x^l, label^l) serve as training data pairs and contribute to the network training process. Fig. 2 provides an overview of the proposed method, which mainly consists of a global context extraction network A_1, a local feature extraction network A_2, and a module for fusing global–local context information.

The primary objective of A_1 is to extract useful global contextual information. A deep learning model is trained to filter out this useful information

$$f_{\text{global}} = E_1\left(x^l \times G\left(E_1\left(x^l\right)\right)\right) \tag{1}$$

where f_{global} represents the extraction of global context features. E_1 is the feature encoder of the auxiliary branch,

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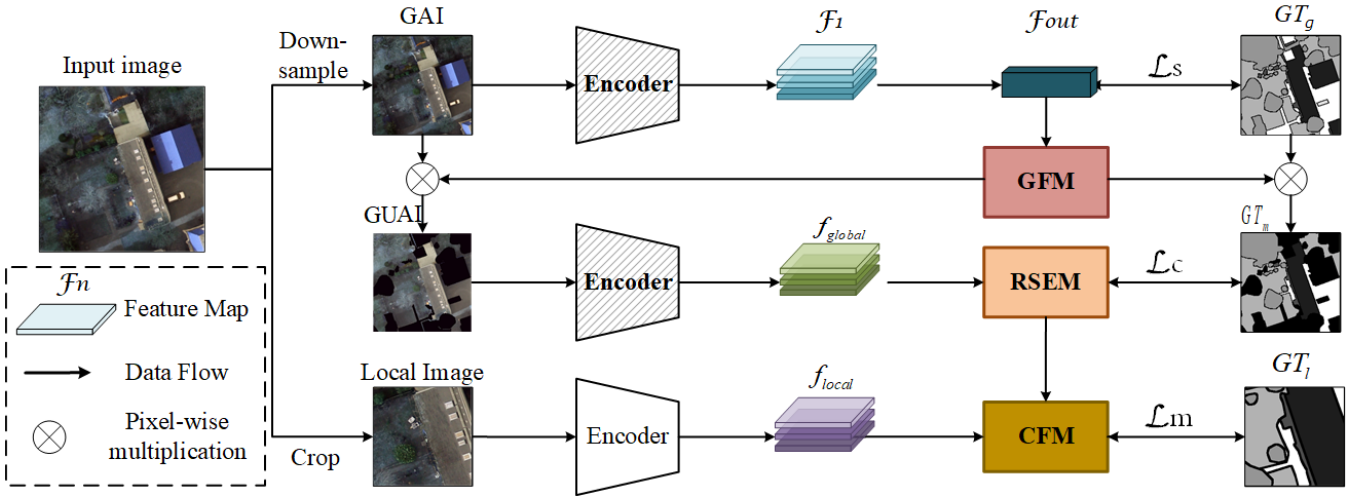

Fig. 2. Architecture of our proposed network consists of three main components: useful global context extraction network, local context extraction network, and feature fusion module. Global context extraction network is represented by the first and second rows in the figure and comprises three parts: explicitly GFM, feature extraction module, and RSEM. The third row includes local feature extraction and global–local CFMs, which are used to generate final segmentation results during testing.

and $G(\cdot)$ represents the spatial position relationship used for obtaining useful information. The symbol "×" signifies pixel-by-pixel multiplication.

For A_2, local feature extraction occurs using (x^s, label^s) as the input to E_2, which yields the local context information f_{local}

$$f_{\text{local}} = E_2\left(x^s\right). \tag{2}$$

The output $f_{\text{local}} \in \mathbb{R}^{c_m \times h_m \times w_m}$ from the feature encoder E_2, and c_m, h_m, w_m corresponds to the channel number, height, and width of the final layer feature map, respectively.

Feature fusion is performed between local features f_{local} and global features f_{global} to obtain the final feature map f_{final}. Importantly, before this fusion, related-category semantic enhancement is applied to the global features

$$f_{\text{final}} = F\left(f_{\text{local}}, H\left(f_{\text{global}}\right)\right). \tag{3}$$

The output $f_{\text{final}} \in R^{c_m \times h_m \times w_m}$ comes from the auxiliary feature fusion network, where $F(\cdot)$ represents the feature fusion network and $H(\cdot)$ represents related-category semantic enhancement operations. Subsequently, the fused feature map undergoes processing by a 1×1 convolution layer and is then upsampled to obtain the final segmentation map.

B. Explicit Guidance Feature Extraction Module

This study attempts to use a global image to compensate for truncated features in the local image (as depicted in Fig. 1). The objective is to guide the network through continuous training to suppress regions unrelated to the local image and to learn more useful information. Since the GUAI lacks ground truth, we use an auxiliary semantic segmentation network to obtain GAI. This involves filtering out useless information from GAI and masking it to obtain the GUAI. This process is shown in the upper part of Fig. 2 and involves two key steps. First, GAI is input into the auxiliary semantic segmentation network to obtain preliminary rough semantic segmentation results. These results are then processed by the explicit GFM to calculate the useful area for local segmentation in the auxiliary feature map.

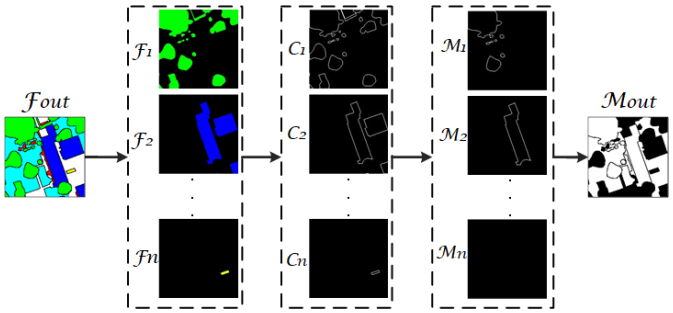

Fig. 3. Explicit GFM.

After downsampling GAI S times, it is input into an auxiliary segmentation network to obtain coarse segmentation results, yielding segmentation results F_c as shown in the following equation:

$$F_c = E_1\left(\text{resize}\left(x^l\right)\right). \tag{4}$$

In this study, the position of the local image within the global image is defined as $P_L = [((W_g - W_l/2S), (H_g - H_l/2S)), ((W_g + W_l/2S), (H_g + H_l/2S))]$, where (W_g, H_g) and (W_l, H_l) represent the widths and heights of the global and local images, respectively, with S indicating the downsampling rate of global image. Essentially, the local image is located in the center of the global image (as shown in Fig. 1). As shown in Fig. 3, to determine the connection of objects in GAI to the local image area, the semantic segmentation result F_c is initially divided into n one-hot map channels $(O = O_1, O_2 \ldots O_n)$, as shown in 5, where n is the total number of classes

$$[O_1, O_2, \ldots, O_n] = \text{Onehot}(F_c). \tag{5}$$

Subsequently, we identify complete object contours $(C_i = [C_i{}^1, C_i{}^2 \ldots C_i{}^k], i = [1, n])$ within the one-hot map O,

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checking each for intersection with the position of the local image. Nonintersecting contours are deemed interfering areas and are masked. The spatial positions of useful areas are then calculated to produce M_{out}

$$M_{\text{out}}(C_i) = \begin{cases} 0, & \text{if } \text{sum}(C_i \cap P_L) = 0 \\ 1, & \text{if } \text{sum}(C_i \cap P_L) > 0. \end{cases} \tag{6}$$

The specific process is shown in Fig. 3. This method screens the spatial positions of all useful areas. The original GAI and label are then multiplied by M_{out} to obtain a global useful auxiliary image (GUAI) and label containing only useful information (7). The GUAI is subsequently input into the feature extraction network for global feature extraction

$$\text{GUAI} = \text{GAI} \times M_{\text{out}}. \tag{7}$$

Significantly, the network is used for useful information screening, sharing weights with the global context feature extraction network. As the network undergoes continuous training, its reliability in extracting useful information improves.

C. Related-Category Semantic Enhancement Module

One challenge in remote-sensing image semantic segmentation is the lower spatial resolution compared with natural images (such as the Gaofen-2 satellite images with 1-m panchromatic/4-m multispectral resolution [50]). The loss of spatial detail is exacerbated during feature encoding. In this study, the downsampled GAI is input into the network to obtain global context information, which further blurs spatial details of GAI and poses challenges to global feature extraction. To overcome this, we designed a network that reduces the semantic distance between pixels of related categories in the feature map, mitigating the impact of spatial detail loss from downsampling. Consequently, we propose RSEM, which enhances related categories by spatially aggregating high-level feature maps. The process involves the following specific steps.

First, the feature map $F_{\text{global}} \in \mathbb{R}^{c_m \times h_m \times w_m}$, containing only useful information, undergoes two convolution operations to generate feature maps ${F_2}^a \in \mathbb{R}^{(c_m/n) \times h_m \times w_m}$ and ${F_3}^a \in \mathbb{R}^{n \times h_m \times w_m}$. Here, n represents the total number of categories. To ensure accurate category prediction, we generate a new ground truth (as shown in (8) which is represented by GT_m in Fig. 2) by processing the labels of the training data, which is then downsampled to match the size of the feature map. This downsampled ground truth supervises ${F_3}^a$ for the useful auxiliary feature map label

$$GT_m = GT_g \times M_{\text{out}}. \tag{8}$$

For spatial positioning of each category, we one-hot encode the classification results to generate $[M_1 \ldots M_n]$ (as per 9). Multiplying $M_i, i = [1, n]$ by ${F_2}^a$ yields a category-specific feature map

$$[M_1, M_2, \ldots, M_n] = \text{Onehot}\left(F_3^a\right) \tag{9}$$

$$[R_1, R_2, \ldots, R_n] = [M_1, M_2, \ldots, M_n] * F_2^a. \tag{10}$$

Finally, these category-specific feature maps are concatenated in sequence (11) and serve as the global feature map input for the subsequent layer

$$F_{\text{out}} = \text{Cat}(\text{repeat}(R_1), \text{repeat}(R_2), \ldots, \text{repeat}(R_n)). \tag{11}$$

Fig. 4 illustrates the related-category semantic enhancement process, which aims to minimize semantic distances between pixels of the same category while maximizing those between different categories. This enhances the feature fusion capability of the network. Therefore, this process is applied only before feature fusion, specifically for the deepest layer of the auxiliary feature map.

D. Global–Local Contextual Cross-Fusion Module

The fusion of global and local contexts is a crucial next step. Inspired by the effective feature interaction facilitated by cross-attention, we designed a global–local contextual CFM. This module includes self-attention and cross-attention components, with cross-attention transferring information from global to local feature maps. Instead of directly adding global feature map information to the local image, we project a biased focus to enhance the features of the local feature map.

Before feature fusion, to account for long-range dependencies in the feature map, we separately process the global useful auxiliary feature map and main segmentation feature map through self-attention (as shown in the right-hand side of Fig. 5)

$$y_i = \frac{1}{\text{NF}(x)} \sum_{\forall j} f\left(x_i, x_j\right) g\left(x_j\right) \tag{12}$$

where i is the output position index, j enumerates all possible positions in x, $f(x_i, x_j)$ computes the similarity matrix between x_i and its contextual feature x_j, $g(x_j)$ calculates the representation of the feature at position j, and $\text{NF}(x)$ is the normalization factor. This equation is then processed through a residual connection block to obtain z, completing the self-attention module (as shown in the left-hand side of Fig. 5).

Drawing inspiration from OCRnet [51], we use cross-attention for information interaction in the dual-branch network (right-hand side of Fig. 5). Cross-attention is calculated by obtaining queries from one modality and keys and values from another. We transform the spatial features of the auxiliary feature layer into keys K_g and values V_g and the spatial features of the local feature layer into queries Q_l. The specific calculation process is as follows:

$$H_g = \text{Softmax}\left(Q_l K_g^T / \sqrt{d}\right) V_g \tag{13}$$

where H_g represents the attention map and d signifies a custom constant; Q_l denotes the linear projection of feature of local image; and K_g and V_g are two linear projections of GUAI feature.

E. Loss Function

During training, the entire network is trained using both the main and auxiliary segmentation losses. The loss function of the network is a combination of these two losses

$$\mathcal{L} = \mathcal{L}_m + \mathcal{L}_a. \tag{14}$$

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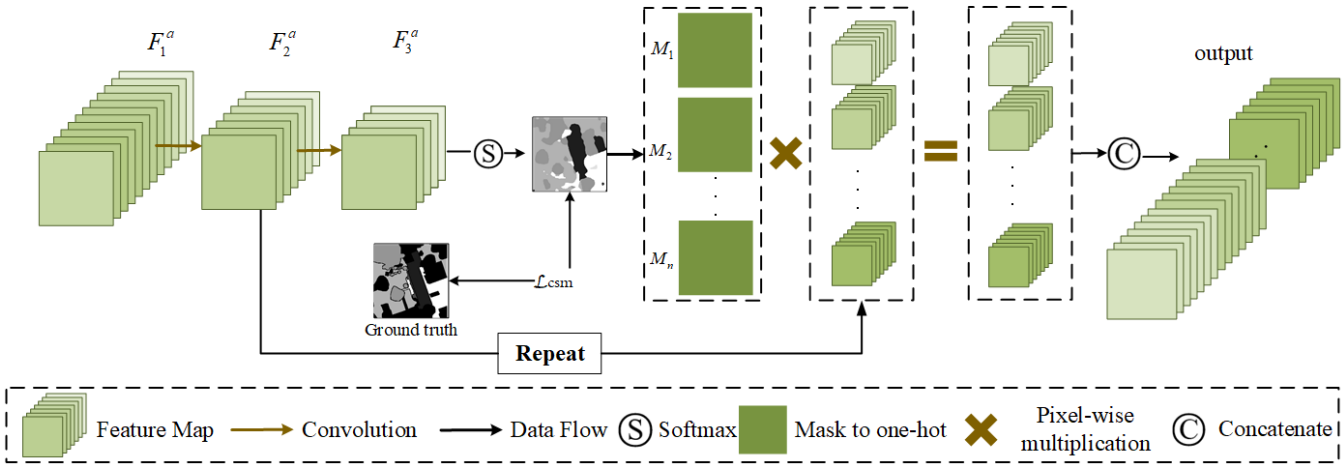

Fig. 4. Structure of RSEM. Ground truth is obtained by downsampling GT_m in Fig. 2.

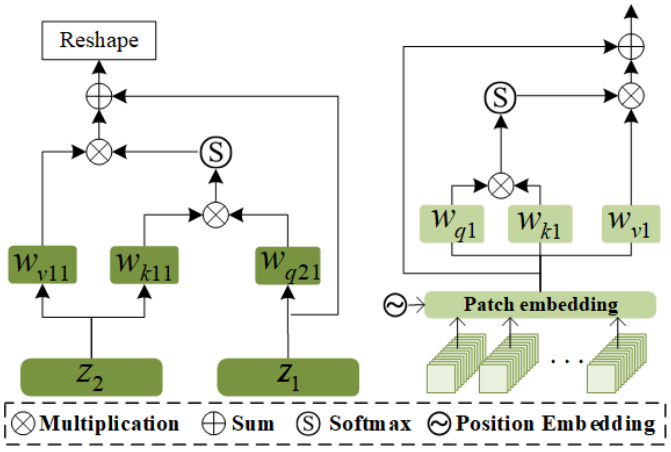

Fig. 5. Flowchart of global–local context information fusion module. The right side represents the self-attention mechanism. Both the local feature map and the global feature map need to be enhanced by the self-attention mechanism. The left side represents cross attention, and z_1 and z_2 represent the enhanced local feature map and the global feature map, respectively.

The main segmentation loss $\mathcal{L}_m$ computed using multiclass cross-entropy loss

$$\mathcal{L}_m = -\frac{1}{M}\sum_{i=1}^{n}\sum_{c=1}^{n} y_{ic}\log(p_{ic}). \tag{15}$$

In this formula, M and n denote the numbers of samples and classes, respectively. For example, y_{ic} is 1 if its true category is c; otherwise, it is 0. p_{ic} is the probability that sample i belongs to category c. The auxiliary segmentation loss comprises the intraclass feature enhancement loss $\mathcal{L}_{sse}$ and segmentation loss $\mathcal{L}_s$

$$\mathcal{L}_a = \mathcal{L}_{sse} + \mathcal{L}_s. \tag{16}$$

IV. Experimental Results

We conduct a comprehensive evaluation of our proposed network using the GID, BLU, and ISPRS Potsdam datasets. This section provides an overview of these datasets and specifics of our implementation, followed by a series of ablation studies to investigate the functionalities of individual network modules. Finally, we compare the results our method with those of other methods on these datasets.

A. Dataset

The GID dataset, released by Wuhan University, is a remote-sensing land cover dataset based on high-resolution Gaofen-2 satellite data [52]. This dataset comprises ten remote-sensing images, each measuring 7200 × 6800 pixels with a 1-m panchromatic/4-m multispectral spatial resolution. It encompasses 16 land cover classes such as industrial land (IDL), urban residential (UR) and rural residential (RR) areas, traffic land (TL), paddy fields (PFs), irrigated land (IL) and dry croplands (DCs), garden plots (GPs), arbor woodland (AW), shrub land (SL), natural grassland (NG), artificial grassland (AG), river (RV), lake (LK), and pond (pn). For our study, we divided each image into 16 smaller images of 2048 × 2048 pixels. These subimages were then randomly allocated into training (90 subimages), validation (30 subimages), and test sets (40 subimages).

The BLU dataset, released by the Information Engineering University of the Strategic Forces of the People's Liberation Army (PLA), is based on high-resolution satellite data from the Beijing-2 satellite. This dataset contains four extensive remote-sensing images, each with a size of 15 680 × 15 680 pixels and a spatial resolution of 0.8 m. It categorizes land cover into six types: background (BG), building (BD), vegetation (V), water (W), farmland (F), and road (R).

The ISPRS Potsdam dataset includes 38 high-resolution aerial images, each measuring 6000 × 6000 pixels with a 5-cm spatial resolution. These images feature four channels: near-infrared, red, green, and blue, and encompass six land cover types: impervious surfaces (I), buildings (B), low vegetation (L), trees (T), cars (C), and background. In our study, 18 images were used for training, six for validation, and 14 for testing, following the allocation of training, validation, and testing sets is consistent with [53], [54], [55], and [56].

B. Implementation Details

For the GID and BLU datasets, each image was subjected to a sliding cropping process to produce multiple

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2048 × 2048 pixel subimages. Before being fed into the network, 100 local images were randomly cropped from each subimage. In contrast, for the Potsdam dataset, with its larger 6000 × 6000 pixel images, 1000 local images were randomly cropped for each image prior to network input. We used a 256 × 256 pixel window for the random cropping of local images. Correspondingly, GUAIs were cropped to 768 × 768 pixels and downsampled 4 × to generate training samples. This cropping and downsampling approach for the GUAIs is further explained in the ablation study. For the GID and BLU datasets, each image is initially slidingly cropped into multiple subimages of dimensions 2048 × 2048 pixels. On being fed into the network, 100 local images are randomly cropped for every subimage; in the case of Potsdam dataset, the image size is 6000 × 6000. On being fed into the network, 1000 local images are randomly cropped for every image. During inference, the same sliding method was used to crop input image blocks from the original test images, with each block being processed separately. To evaluate the performance of our method, we use three evaluation metrics: mIOU, F1 score, and OA.

The mIOU computes the average intersection-over-union for each class in the dataset

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{17}$$

$$\text{mIOU} = \frac{\sum_{i=1}^{n} \text{IOU}_i}{n}. \tag{18}$$

The F1 score is the harmonic mean of precision and recall

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{20}$$

OA considers the overall classification accuracy. These metrics are calculated using the counts of true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs) pixels as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{21}$$

In our clustered learning method, training spanned 50 epochs, using a stochastic gradient descent optimizer. The base learning rate was set to 0.01, with a reduction applied after each iteration according to the formula $[1 - (\text{iter}/\text{total}_i\text{ter})]^{0.9}$. Both the momentum decay coefficient and weight decay coefficient were set at 0.9 and 0.00001, respectively. The network was developed using PyTorch and trained on a TITANRTX × 24 GB GPU.

C. Ablation Study

To investigate the impact of each component in our approach, we perform ablation experiments comparing five models: FCN, FCN + CFM, FCN + CFM + GFM, FCN + CFM+RSEM, and FCN+CFM+GFM+RSEM. The baseline FCN model did not include these three components. These experiments were conducted on the GID, BLU, and Potsdam datasets, with all the models using a pretrained ResNet50 backbone on ImageNet.

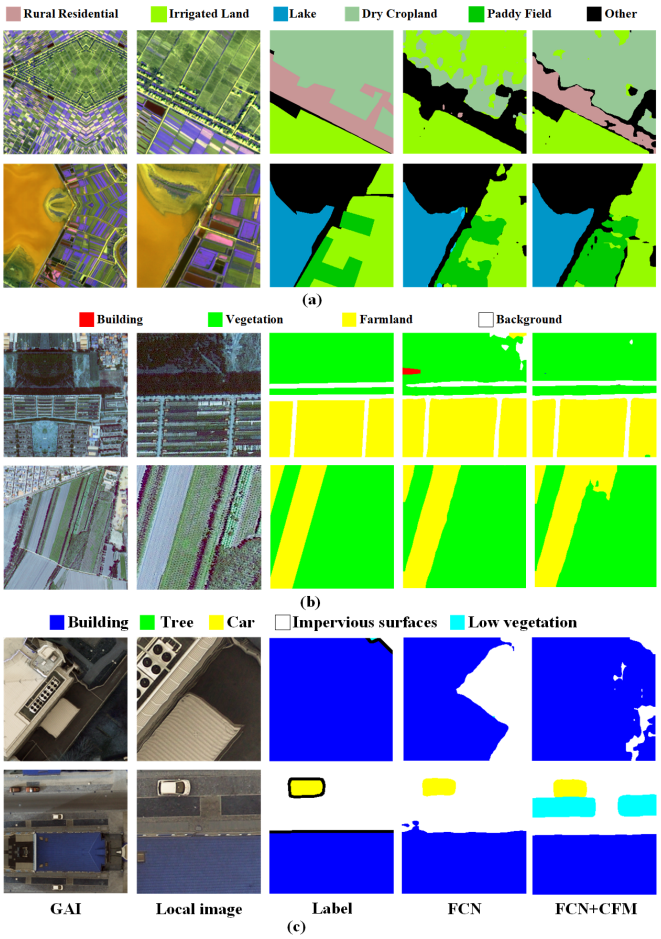

Fig. 6. Example results of different semantic segmentation models. Columns from (left) to (right) show global images, local images, ground truth, results of FCN model, and results of FCN + CFM, respectively. (a) GID. (b) BLU. (c) Potsdam.

1) Effect of Global–Local Context Information Fusion Module: As shown in Table I, applying a dual-branch network for global context information fusion on the Potsdam dataset resulted in an increased segmentation accuracy. Specifically, there was an increase of 0.61%, 0.53%, and 0.48% in OA, MIoU, and Ave.F1, respectively. Notably, the "Impervious surface" category saw the most significant improvement with a 1.36% increase in IoU. However, there were slight declines in the "tree" and "car" categories. The GID dataset showed even more significant improvements, as shown in Table II, with an overall accuracy increase of 1.43%. However, categories such as GP, NGs, and LKs experienced decreases. The BLU dataset followed a similar pattern, with increases of 0.55%, 0.39%, and 0.66% in OA, MIoU, and Ave.F1, respectively, but a significant decrease in the category of "road." Visual comparisons of semantic segmentation results are shown in Fig. 6. The first row of Fig. 6(a)–(c) illustrates that feature fusion effectively enhances accuracy in some categories. However, the second row highlights that direct input of entire images can introduce noise, resulting in errors due to complex backgrounds, emphasizing the need to remove background noise interference in remote-sensing image segmentation.

2) Effect of Explicit GFM: As shown in Tables I–III, the explicit GFM improved semantic segmentation results

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TABLE I
ABLATION EXPERIMENTS ON THE POTSDAM DATASET

BaseModel	model			IOU					Evaluation index		
	CFM	GFM	SSEM	I	B	L	T	C	OA	mF1	mIou
FCN				85.37	93.32	74.96	75.29	89.98	89.37	90.99	83.78
	+			86.73	**94.39**	75.41	75.06	89.96	89.98	91.29	84.31
	+	+		86.59	93.97	76.37	75.83	90.47	90.1	91.51	84.65
	+		+	**87.01**	94.4	76.17	75.47	90.45	90.19	91.53	84.7
	+	+	+	86.74	94.2	**76.75**	**76.66**	**90.63**	**90.28**	**91.73**	**85**

TABLE II
ABLATION EXPERIMENTS ON THE GID DATASET

Basemodel	model			IoU																Evaluation Index		
	CFM	GFM	SSEM	OT	IDL	UR	RR	TL	PF	IL	DC	GP	AW	SL	NG	AG	RV	LK	PN	OA	mF1	mIou
FCN				79.51	51.3	58.17	50.91	50.31	46.95	71.67	12.06	6.05	76.49	4.21	63.72	15.81	73.21	45.76	22.38	84.4	59.03	47.53
	+			81.41	59.69	62.38	54.08	55.75	56.03	76.3	60.35	4.98	78.11	8.91	63.68	**17.61**	78.52	44.84	**29.4**	85.73	64.59	52.00
	+	+		81.22	**60.25**	**62.59**	54.14	51.64	58.4	76.69	55.63	**11.76**	**80.53**	5.55	**64.76**	16.25	**82.88**	**54.57**	23.24	85.95	64.91	52.51
	+		+	**81.92**	57.67	61.85	54.41	55.94	**60.42**	**77.6**	**60.93**	8.91	80.4	11.66	62.96	15.65	81.3	44.18	22.59	86.26	64.93	52.4
	+	+	+	81.69	57.7	62.14	**55.03**	**56.2**	59.22	77.22	60.03	9.85	80.23	**12.25**	64.39	16.35	79.92	51.23	23.52	**86.41**	**65.58**	**52.94**

TABLE III
ABLATION EXPERIMENTS ON THE BLU DATASET

BaseModel	model			IoU						Evalution index		
	CFM	GFM	RSEM	BG	BD	V	W	F	R	OA	mF1	mIou
FCN				57.07	77.86	82.43	73.86	76.1	53.04	86.51	81.88	70.06
	+			**58.02**	78.64	83.34	75.36	77.51	51.48	87.06	82.27	70.72
	+	+		57.58	78.57	83.33	**76.89**	78.08	53.05	87.18	82.66	71.25
	+		+	57.89	78.39	83.16	75.44	77.56	53.77	87.16	82.55	71.04
	+	+	+	57.75	**79.21**	**83.47**	76.18	**78.29**	**54.61**	**87.42**	**82.93**	**71.58**

compared with the FCN + CFM configuration. For the Potsdam dataset, there was an overall accuracy improvement of 0.12%, an increase of 0.22% in the avg. F1 score, and a 0.34% rise in MIoU. For the GID dataset, the overall accuracy improved by 0.22%, the avg. F1 score by 0.32%, and MIoU by 0.51%. The BLU dataset saw increases of 0.12% in overall accuracy, 0.39% in avg. F1 score, and 0.53% in MIoU. Compared with the baseline FCN, each category on the three datasets showed some improvement, addressing the problem of accuracy decline in some categories after introducing global context. Fig. 7 shows segmentation results with the implemented feature constraints. Taking the Potsdam dataset as an example, the "tree" in the lower right corner was challenging to segment due to insufficient global context, as shown in the third line of Fig. 7. Direct fusion of the global auxiliary context led to misclassification of a "car" as a "tree" due to noise. However, when applying explicit guidance feature extraction to GAI, the network effectively segmented both "car" and "tree." The experiments demonstrate that selectively fusing global features can effectively reduce negative impacts from GAI and improve segmentation accuracy for highly similar objects.

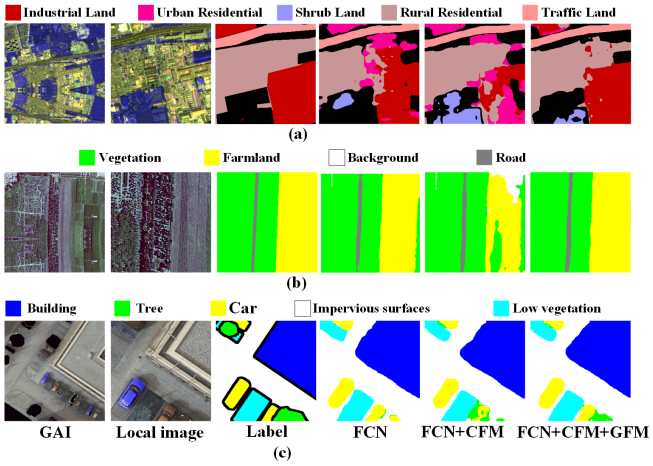

Fig. 7. Example results of different semantic segmentation models. Columns from (left) to (right) show global images, local images, ground truth, results of FCN model, results of FCN + CFM, and results of FCN + CFM + GFM, respectively. (a) GID. (b) BLU. (c) Potsdam.

3) Effect of the RSEM: This module enhances the semantic information of globally useful auxiliary feature maps. With the inclusion of this module, segmentation accuracies on the

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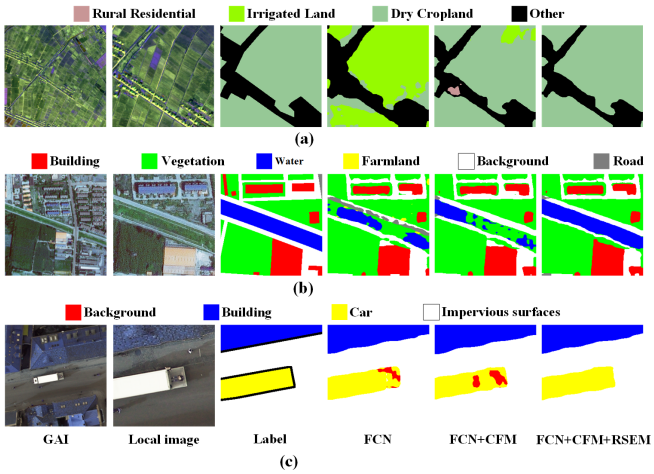

Fig. 8. Example results of different semantic segmentation models. Columns from (left) to (right) show global images, local images, ground truth, results of FCN model, results of FCN + CFM, and results of FCN + CFM + CSEM, respectively. (a) GID. (b) BLU. (c) Potsdam.

Potsdam, BLU, and GID datasets increased by 0.21%, 0.1%, and 0.53% respectively. When applied together with the GFM module, the network not only filters useful information but also enhances semantic information, leading to more pronounced improvements. For Potsdam, the overall accuracy improved by 0.30%, mF1 by 0.44%, and mIOU by 0.69%. For BLU, overall accuracy increased by 0.36%, mF1 by 0.66% and mIOU by 0.86%. For GID, improvements were observed with a 0.68% increase in overall accuracy, 0.99% in mF1, and 0.94% in mIOU. Fig. 8 shows segmentation results after introducing the relevant category semantic enhancement module. These results indicate that the module provides a more accurate global context for local segmentation, thereby improving segmentation accuracy. Fig. 8 shows the segmentation results after introducing the related category semantic enhancement module. The experimental results demonstrate that this module can provide a more accurate global context for local segmentation, thereby improving segmentation accuracy.

4) Effect of GAI Size: The size of GUAI is crucial in determining the scope of context information modeled. To adequately compensate for the truncated geographic information, the GUAI should be l times larger than the local image. However, implementing self-attention and cross-attention during feature fusion could diminish local information in an excessively large auxiliary feature map. Thus, we tested different GUAI sizes in our experiments. As shown in Table IV, the best results on the GID and BLU datasets were achieved with $l = 3$, while for the Potsdam dataset, $l = 4$ was optimal. However, the improvements over $l = 3$ were marginal. Consequently, $l = 3$ was selected for this study based on overall performance.

5) Effect of Downsampling Rate on GAIs: The downsampling rate s directly affects the contextual detail represented in the auxiliary map. Additional experiments varying this rate showed that performance declines with increasing downsampling rates. As shown in Table V, optimal results were achieved at $s = 3$ for all the datasets, though this significantly slowed segmentation speed. Therefore, balancing segmentation accuracy and speed, we opted for $s = 4$ in our experiments.

TABLE IV
EFFECT OF AUXILIARY IMAGE SIZE

DataSet	L	OA	Avg. F1	mIou	Time/per epoch
GID	2	86.17	65.32	52.55	5'32"
	3	**86.41**	65.58	**52.94**	8'03"
	4	86.35	**65.73**	52.87	11'13"
BLU	2	87.09	82.31	70.96	4'18"
	3	**87.42**	82.93	71.58	5'39"
	4	87.28	**83.16**	**71.78**	9'45"
Potsdam	2	89.93	91.31	84.36	7'31"
	3	90.28	91.73	85	10'08"
	4	**90.31**	**91.85**	**85.27**	15'47"

TABLE V
EFFECT OF DOWNSAMPLING RATE ON AUXILIARY IMAGES

DataSet	s	OA	Avg. F1	mIou	Time/per epoch
GID	3	**86.58**	**65.81**	**53.16**	9'52"
	4	86.41	65.58	52.94	8'03"
	6	85.87	64.66	52.07	5'17"
BLU	3	**87.52**	**82.95**	**71.73**	8'48"
	4	87.42	82.93	71.58	5'39"
	6	86.69	82.06	70.84	4'03"
Potsdam	3	**90.29**	**91.88**	**85.17**	13'25"
	4	90.28	91.73	85	10'08"
	6	90.02	91.31	84.72	6'40"

D. Comparison With Popular Methods

To establish the superiority of the proposed TCNet, it was compared with several state-of-the-art semantic segmentation methods. These included classic networks such as FCN, DeepLab V3+ [30], and PSPNet [34], single-branch context aggregating networks such as ST-UNet [16], BiTSRS [57], CF-Net [58], and LANet [59], as well as dual-branch context aggregating networks such as GLNet [21] and WiCoNet [49]. Some authors have publicly released the source codes in their respective papers. For networks with available source code, we retrained them using the provided code. For those without accessible source code, we relied on the experimental published by the authors.

1) Results on the GID Dataset: Table VI presents the performance of each semantic segmentation method. TCNet outperformed others in OA, MIoU, and Ave.F1. DeepLabV3+, with its ASPP module and unique decoder structure, outperformed other DCNN-based models. ST-UNet, leveraging the Swin Transformer for global contextual modeling, showed slightly better results than CF-Net and LANet, which expands the acceptance field of low-level feature maps through channel attention or local attention based on high-level features. WiCoNet models' bilateral semantic dependency using a contextual transformer achieved superior segmentation performance compared with GLNet, another bilateral network. TCNet surpassed DeepLab V3+ with a 1.47% improvement in OA, 2.97% in MIoU, and 3.85% in Ave.F1. It also outperformed ST-UNet with a 2.65% increase in MIoU and a 2.28% rise in Ave.F1. Compared with WiCoNet, TCNet showed

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TABLE VI
COMPARATIVE EXPERIMENT WITH THE GID DATASET

Model	Iou																Evaluation Index		
	OT	IDL	UR	RR	TL	PF	IL	DC	GP	AW	SL	NG	AG	RV	LK	PN	OA	mF1	mIou
FCN	79.51	51.3	58.17	50.91	50.31	46.95	71.67	12.06	6.05	76.49	4.21	63.72	15.81	73.21	45.76	22.38	84.4	59.03	47.53
Deeplabv3+	80.94	56.48	61.36	51.34	50.26	51.14	71.93	56.31	9.61	76.52	11.04	64.02	14.42	78.52	43.87	21.81	84.94	61.73	49.97
PSPNet	78.91	54.42	57.34	51.6	48.21	50.55	75.94	56.45	7.71	75.93	5.29	63.33	14.58	75.84	41.6	25.82	84.67	60.84	48.97
ST-UNet	80.53	56.91	59.49	53.74	49.34	55.71	75.29	57.86	15.7	76.86	4.6	58.57	15.55	79.48	38.4	26.59	-	63.3	50.29
CF-Net	74.78	55.86	59.42	49.13	52.64	50.39	77.08	51.4	4.57	75.87	5.65	65.29	13.15	77.61	40.93	22.77	84.84	61.34	48.6
LANet	78.74	47.64	58.49	51.53	45	34.35	72.57	51.95	1.37	74.88	2.12	64.27	14.98	76	48.31	21.02	83.25	60.29	48.01
GLNet	75.13	38.67	59.3	39.36	48.16	59.04	72.11	38.52	1.09	64.28	9.96	**69.89**	11.88	74.21	**53.05**	**29.02**	-	59.74	46.48
WiCoNet	81.24	**57.84**	61.35	54.43	54.34	56.49	76.83	59.74	7.03	**79.13**	1.16	62.12	12.42	**80.49**	46.51	27.62	85.79	63.27	51.17
Ours	**81.69**	57.7	**62.14**	**55.03**	**56.2**	**59.22**	**77.22**	**60.03**	**9.85**	78.94	**12.25**	64.39	**16.35**	79.92	51.23	24.81	**86.41**	**65.58**	**52.94**

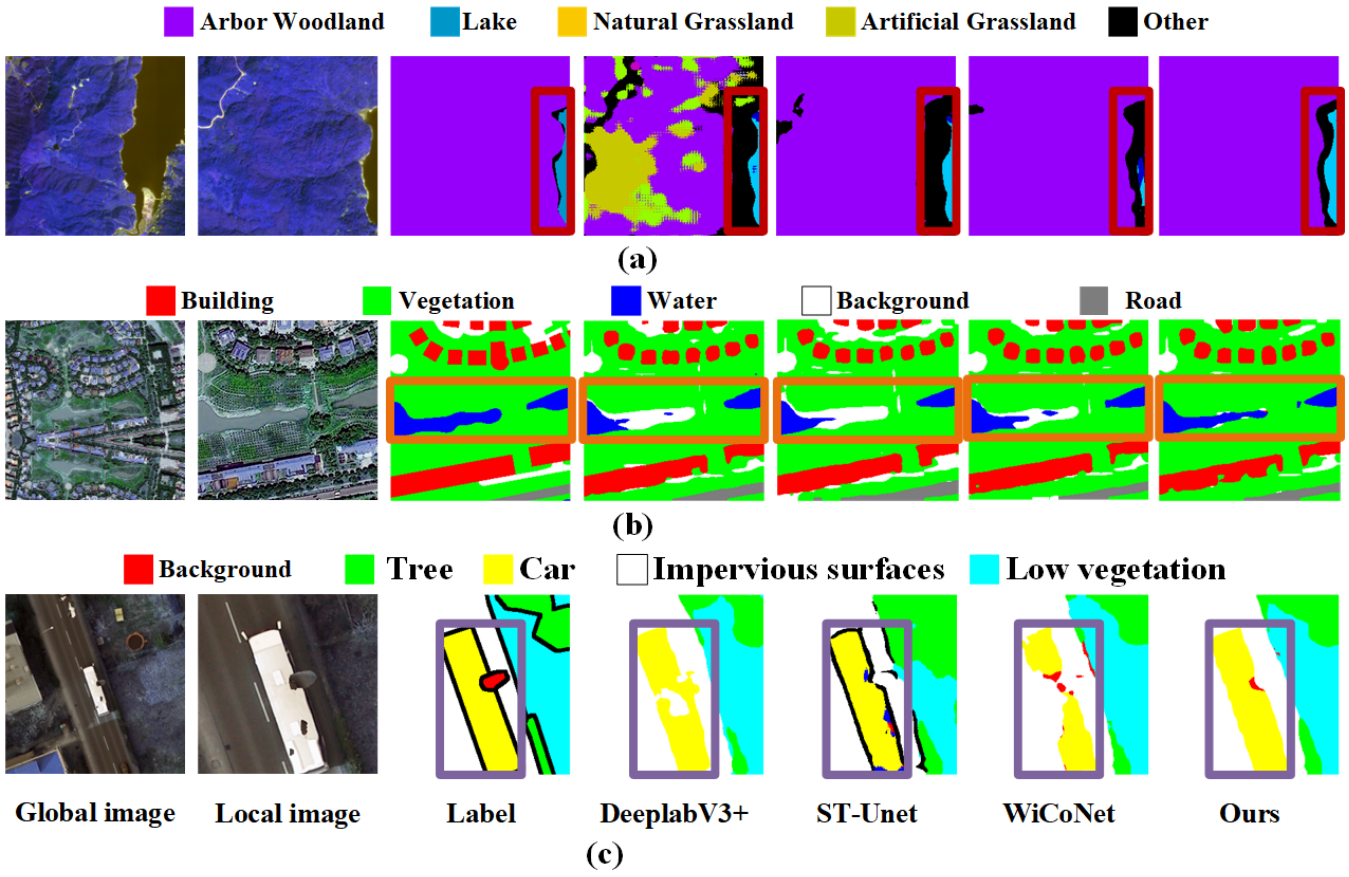

Fig. 9. Example results of different semantic segmentation models. Columns from (left) to (right) show images, labels, results of DeepLabV3+ model, results of ST-UNet model, results of WiCoNet model, and results of our model, respectively. (a) GID. (b) BLU. (c) Potsdam.

improvements of 0.62% in OA, 1.77% in MIoU, and 2.31% in Ave.F1. These outcomes indicate that while independent extraction of global contextual information is useful, incorporating it into local image segmentation, as done in GLNet and WiCoNet, is not sufficient for remote-sensing images. The superiority of TCNet over these methods demonstrates the effectiveness of our approach in filtering global information.

2) Results on the BLU Dataset: Table VII details the performance of all the tested CNNs on the BLU dataset. Across all the evaluation settings, TCNet emerged as the most accurate. Compared with WiCoNet, TCNet showed improvements in OA, avg. F1 score, and mIOU by 0.43%, 0.42%, and 0.56% respectively.

Using DeepLabV3+, ST-UNet, and WiCoNet as examples, Fig. 9 shows comparison of the segmentation effects of TCNet against these models. The data in the second row of the figure were derived from the BLU dataset. As highlighted by the orange box, DeepLab and ST-UNet networks struggled with segmenting water bodies due to insufficient global context information. WiConet, despite introducing global context, encounters challenges with complex backgrounds, impacting its ability to segment waster bodies in this instance. However,

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TABLE VII
COMPARATIVE EXPERIMENT WITH THE BLU DATASET

Model	IoU						Evaluation index		
	BG	BD	V	W	F	R	OA	mF1	mIou
FCN	57.07	77.86	82.43	73.86	76.1	53.04	86.51	81.88	70.06
DeepLabV3+	54.71	75.91	82.48	68.37	77.41	50.08	86.26	80.44	68.16
PSPNet	56.58	78.01	82.13	**76.22**	76.47	53.01	86.65	82.24	70.4
ST-UNet	54.14	76.47	81.78	72.89	76.3	47.73	-	80.4	68.22
CF-Net	**57.91**	78.84	82.72	75.96	76.89	52.31	86.87	82.31	70.74
LANet	57.02	77.56	82.51	75.16	76.61	53.61	86.52	82.07	70.4
GLNet	57.62	78.35	83.17	73.75	77.62	52.58	-	82.16	70.51
WiCoNet	57.61	78.83	83.13	75.91	77.65	52.98	86.99	82.51	71.02
ours	57.75	**79.21**	**83.47**	76.18	**78.29**	**54.61**	87.42	82.93	71.58

TABLE VIII
COMPARATIVE EXPERIMENT WITH THE POTSDAM DATASET

Model	Iou					Evaluation Index		
	I	B	L	T	C	OA	mF1	mIou
FCN	82.37	93.32	74.96	75.29	89.98	89.37	90.99	83.78
Deeplabv3+	85.88	93.83	75.03	75.51	89.45	89.67	91.09	83.94
PSPNet	84.17	90.36	71.48	75.84	89.61	89.25	89.98	82.29
ST-UNet	**87.69**	93.13	76.36	74.87	90.27	-	90.13	84.46
SwinB-CNN+BD$_{1,2,3}$	-	-	-	-	-	90.0	91.6	-
BiTSRS	84.67	90.86	74.63	75.44	88.11	83.57	-	75.5
UnetFormer	85.21	92.74	79.18	76.87	88.92	89.72	91.58	84.59
GLOTS	85.20	92.19	75.23	76.97	84.37	-	-	82.79
CF-Net	86.81	90.25	**76.76**	76.35	90.42	89.58	90.85	84.12
LANet	85.48	93.79	75.75	75.06	88.51	89.47	90.73	83.72
GLNet	86.12	94.05	74.8	75.87	89.11	-	90.37	83.99
WiCoNet	86.49	**94.31**	75.27	74.18	90.31	89.84	91.16	84.11
Ours	86.74	94.2	76.75	**76.66**	**90.63**	**90.28**	**91.73**	**85.00**

TCNet, augmented by the GFM module, effectively filtered out useless information, resulting in more accurate segmentation of water bodies.

3) Results on the Potsdam Dataset: Table VIII shows the segmentation results for each method on the Potsdam dataset, further supporting TCNet's effectiveness. Our TCNet achieved an OA of 90.28%, MioU of 85.00%, and Ave.F1 of 91.73%, surpassing other methods.

The first and last rows represent results from the GID and Potsdam datasets, respectively. Notably, as indicated by the red box in the first row and purple box in the third row, the introduction of GFM and RSEM in TCNet focuses on leveraging useful information from the global image, enhancing its details. This enhancement is evident in our segmentation results, showcasing superior performance compared with other methods. These comparative experiments confirm the effectiveness of TCNet in semantic segmentation of large-scale remote-sensing images.

E. Scalability

Scalability refers to the ability segmentation of a network to adapt to different image segmentation tasks. In real-world applications, semantic segmentation networks designed for high-resolution images often require retraining for different data sources, which can limit their generalizability. Addressing scalability is a key challenge, especially for real-world applications. Enabling a network to learn more, especially more useful information, compensates for truncated semantic content and

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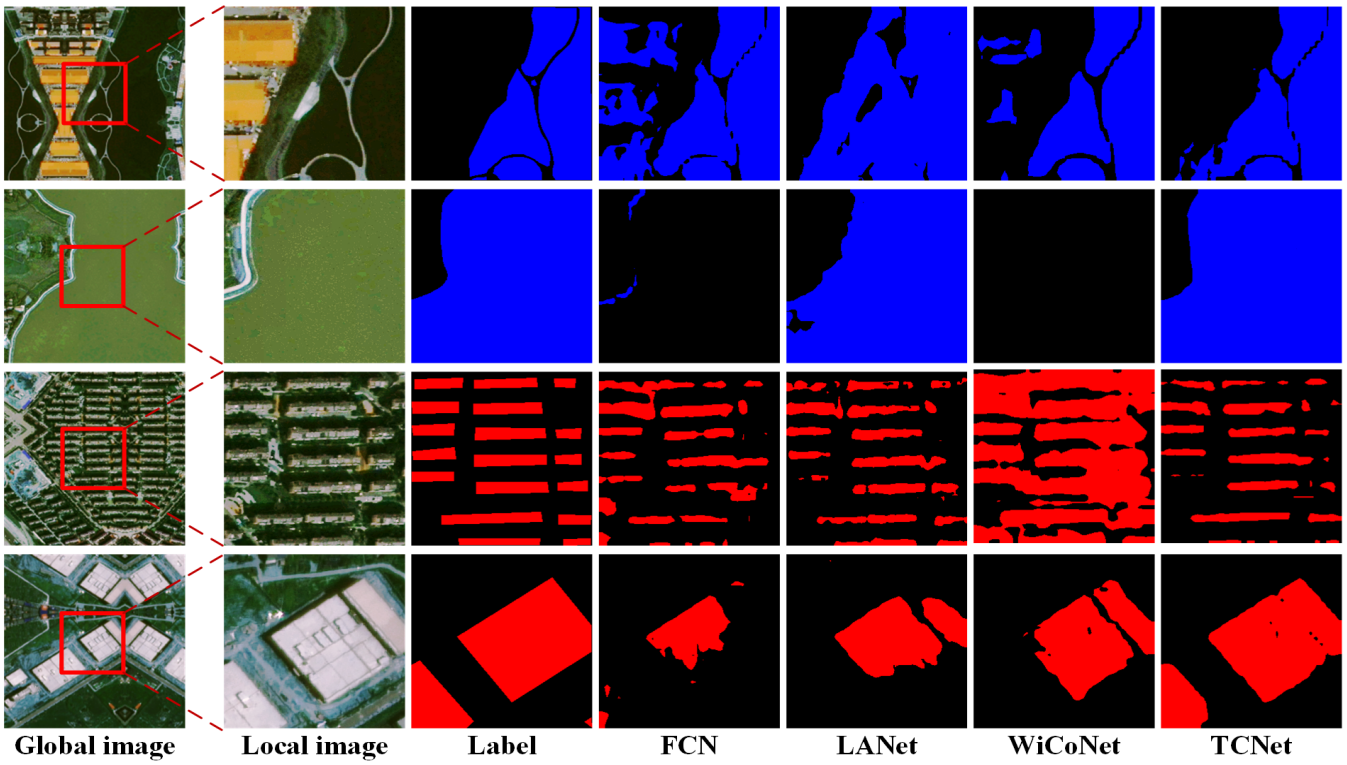

Fig. 10. Semantic segmentation results on Google Earth data.

enhances its generalization ability across different scenarios. Fig. 10 shows our approach with data of 1-m resolution from Google Earth, including meticulously labeled water and buildings. Since these data matched the resolution of the BLU dataset, we use models trained on BLU (FCN, LANet, WiCoNet, and TCNet) to predict water and buildings from Google Earth data. The extraction results reveal that TCNet, compared with alternative methods, demonstrates superior generalization capabilities and performs better in real-world scenarios.

V. Conclusion

In our study, we proposed a truncation compensated semantic segmentation network, which aims to enhance the extraction and utilization of useful information from global contextual features, thereby improving the segmentation of high-resolution scenes. TCNet improves useful information through two ways. First, it implements an explicit guidance mechanism in the feature extraction process, selectively focusing on useful information within the feature map. Second, it uses advanced feature mapping techniques to improve semantic information within specific classes and refine feature details. Furthermore, the network uses a feature fusion module that synergistically combines global and local contextual information. This integrated approach allows TCNet to filter and use both global context and local fine-grained structures more effectively, without incurring additional computational costs. To validate the effectiveness of TCNet's architecture, we conducted comprehensive ablation studies. These studies involved assessing the contribution of each component of TCNet to the overall performance. Our experimental results, derived from evaluations on the GID, BLU, and ISPRS Potsdam datasets, demonstrate that TCNet significantly outperforms the existing benchmark methods in semantic segmentation.

References

[1] C. Zhang, P. A. Harrison, X. Pan, H. Li, I. Sargent, and P. M. Atkinson, "Scale sequence joint deep learning (SS-JDL) for land use and land cover classification," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111593.

[2] J. Zhu et al., "Urban traffic density estimation based on ultrahigh-resolution UAV video and deep neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4968–4981, Dec. 2018.

[3] T. Chowdhury and M. Rahnemoonfar, "Attention based semantic segmentation on UAV dataset for natural disaster damage assessment," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 2325–2328.

[4] L. Hashemi-Beni and A. A. Gebrehiwot, "Flood extent mapping: An integrated method using deep learning and region growing using UAV optical data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2127–2135, 2021.

[5] A. Samie et al., "Examining the impacts of future land use/land cover changes on climate in Punjab province, pakistan: Implications for environmental sustainability and economic growth," *Environ. Sci. Pollut. Res.*, vol. 27, no. 20, pp. 25415–25433, Jul. 2020.

[6] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 4401120.

[7] W. Liu, A. Rabinovich, and A. C. Berg, "ParseNet: Looking wider to see better," 2015, *arXiv:1506.04579*.

[8] B. A. Johnson and L. Ma, "Image segmentation and object-based image analysis for environmental monitoring: Recent areas of interest, researchers' views on the future priorities," *Remote Sens.*, vol. 12, no. 11, p. 1772, Jun. 2020.

[9] L. Ma, M. Li, Y. Gao, T. Chen, X. Ma, and L. Qu, "A novel wrapper approach for feature selection in object-based image classification using polygon-based cross-validation," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 409–413, Mar. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:50:09 UTC from IEEE Xplore. Restrictions apply.

[10] L. Ma, M. Li, X. Ma, L. Cheng, P. Du, and Y. Liu, "A review of supervised object-based land-cover image classification," *ISPRS J. Photogrammetry Remote Sens.*, vol. 130, pp. 277–293, Aug. 2017.

[11] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[12] Z. Xu, W. Zhang, T. Zhang, and J. Li, "HRCNet: High-resolution context extraction network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 13, no. 1, p. 71, Dec. 2020.

[13] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607713.

[14] W. Zhou, J. Jin, J. Lei, and L. Yu, "CIMFNet: Cross-layer interaction and multiscale fusion network for semantic segmentation of high-resolution remote sensing images," *IEEE J. Sel. Topics Signal Process.*, vol. 16, no. 4, pp. 666–676, Jun. 2022.

[15] Z. Zhang, M. Lu, S. Ji, H. Yu, and C. Nie, "Rich CNN features for water-body segmentation from very high resolution aerial and satellite imagery," *Remote Sens.*, vol. 13, no. 10, p. 1912, May 2021.

[16] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[17] Y. Li et al., "MFVNet: A deep adaptive fusion network with multiple field-of-views for remote sensing image semantic segmentation," *Sci. China Inf. Sci.*, vol. 66, no. 4, pp. 1–14, Apr. 2023.

[18] C. Huynh, A. T. Tran, K. Luu, and M. Hoai, "Progressive semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16750–16759.

[19] Q. Li, W. Yang, W. Liu, Y. Yu, and S. He, "From contexts to locality: Ultra-high resolution image segmentation via locality-aware contextual correlation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7232–7241.

[20] L. Ding, J. Zhang, and L. Bruzzone, "Semantic segmentation of large-size VHR remote sensing images using a two-stage multiscale training architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5367–5376, Aug. 2020.

[21] W. Chen, Z. Jiang, Z. Wang, K. Cui, and X. Qian, "Collaborative global-local networks for memory-efficient segmentation of ultra-high resolution images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8916–8925.

[22] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[23] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Exp. Syst. Appl.*, vol. 169, May 2021, Art. no. 114417.

[24] I. Ulku and E. Akagündüz, "A survey on deep learning-based architectures for semantic segmentation on 2D images," *Appl. Artif. Intell.*, vol. 36, no. 1, Dec. 2022, Art. no. 2032924.

[25] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention—MICCAI*. Munich, Germany: Springer, 2015, pp. 234–241.

[26] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[27] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[28] L. C. Chen, G. Papandreou, and I. Kokkinos, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Jun. 2017.

[29] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[30] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[31] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[32] S. Feng et al., "CPFNet: Context pyramid fusion network for medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 10, pp. 3008–3018, Oct. 2020.

[33] Q. Zhu et al., "A global context-aware and batch-independent network for road extraction from VHR satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 353–365, May 2021.

[34] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2881–2890.

[35] Y. Liu, D. Minh Nguyen, N. Deligiannis, W. Ding, and A. Munteanu, "Hourglass-ShapeNetwork based semantic segmentation for high resolution aerial imagery," *Remote Sens.*, vol. 9, no. 6, p. 522, May 2017.

[36] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet-a: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[37] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[38] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617515.

[39] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[40] R. P. K. Poudel, U. Bonde, S. Liwicki, and C. Zach, "ContextNet: Exploring context and detail for semantic segmentation in real-time," 2018, *arXiv:1805.04554*.

[41] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiSeNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. ECCV*, 2018, pp. 325–341.

[42] D. Mazzini, "Guided upsampling network for real-time semantic segmentation," 2018, *arXiv:1807.07466*.

[43] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 3146–3154.

[44] H. Zhang, H. Zhang, C. Wang, and J. Xie, "Co-occurrent features in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 548–557.

[45] Y. Yuan, L. Huang, J. Guo, C. Zhang, X. Chen, and J. Wang, "OCNet: Object context network for scene parsing," 2018, *arXiv:1809.00916*.

[46] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Computer Vision—ECCV*. Glasgow, U.K.: Springer, 2020, pp. 173–190.

[47] F. Zhou, R. Hang, H. Shuai, and Q. Liu, "Hierarchical context network for airborne image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407612.

[48] T. Wu, Z. Lei, B. Lin, C. Li, Y. Qu, and Y. Xie, "Patch proposal network for fast semantic segmentation of high-resolution images," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 12402–12409.

[49] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410313.

[50] Z. Wang, S. Zhang, C. Zhang, and B. Wang, "Hidden feature-guided semantic segmentation network for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603417.

[51] Y. Yuan, X. Chen, X. Chen, and J. Wang, "Segmentation transformer: Object-contextual representations for semantic segmentation," 2019, *arXiv:1909.11065*.

[52] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[53] K. Yue, L. Yang, R. Li, W. Hu, F. Zhang, and W. Li, "TreeUNet: Adaptive tree convolutional neural networks for subdecimeter aerial image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 1–13, Oct. 2019.

[54] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.

[55] B. Yu, L. Yang, and F. Chen, "Semantic segmentation for high spatial resolution remote sensing images based on convolution neural network and pyramid pooling module," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 9, pp. 3252–3261, Sep. 2018.

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[56] K. Nogueira, M. D. Mura, J. Chanussot, W. R. Schwartz, and J. A. Dos Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, Oct. 2019.

[57] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "BiTSRS: A bi-decoder transformer segmentor for high-spatial-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 3, p. 840, Feb. 2023.

[58] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601313.

[59] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

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